A High-School-Friendly Tutorial

Exercise 8.5: The Geometric Mean as a Minimum

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1 What the Exercise Is Asking

The exercise has two connected parts.

1. Prove that the geometric mean of positive numbers

$$a_1, a_2, \ldots, a_n$$

can be obtained by minimizing a certain weighted arithmetic mean.

2. Use that minimum formula to prove that the geometric mean is *superadditive*.

For a positive vector

$$\mathbf{a} = (a_1, a_2, \ldots, a_n),$$

define its geometric mean by

$$\mathrm{GM}(\mathbf{a}) = \left(\prod_{k=1}^{n} a_k \right)^{1/n}.$$

The region in the exercise is

$$D = \left\{ (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n : x_k \geq 0 \text{ for every } k, \ \prod_{k=1}^{n} x_k = 1 \right\}.$$

The first statement to prove is

$$\boxed{\mathrm{GM}(\mathbf{a}) = \min_{\mathbf{x} \in D} \frac{1}{n} \sum_{k=1}^{n} a_k x_k.} \tag{8.33}$$

The second statement, called superadditivity, is

$$\boxed{\left(\prod_{k=1}^{n} (a_k + b_k) \right)^{1/n} \geq \left(\prod_{k=1}^{n} a_k \right)^{1/n} + \left(\prod_{k=1}^{n} b_k \right)^{1/n}.} \tag{SA}$$

Here the two lists are added coordinate by coordinate:

$$\mathbf{a} + \mathbf{b} = (a_1 + b_1, a_2 + b_2, \ldots, a_n + b_n).$$

Positivity assumption. For the main proof, we assume

$$a_k > 0 \quad \text{and} \quad b_k > 0 \qquad (k = 1, 2, \ldots, n).$$

This is the case in which the right-hand side of (8.33) has an actual minimum. Section 8 explains carefully what changes if zero entries are allowed.

2 Reading the Notation

2.1 Sum notation

The symbol

$$\sum_{k=1}^{n} a_k x_k$$

means

$$a_1 x_1 + a_2 x_2 + \cdots + a_n x_n.$$

Therefore

$$\frac{1}{n} \sum_{k=1}^{n} a_k x_k$$

is the arithmetic mean of the n numbers

$$a_1 x_1, a_2 x_2, \ldots, a_n x_n.$$

2.2 Product notation

The symbol

$$\prod_{k=1}^{n} a_k$$

means

$$a_1 a_2 \cdots a_n.$$

For example, when $n = 3$,

$$\left(\prod_{k=1}^{3} a_k \right)^{1/3} = (a_1 a_2 a_3)^{1/3}.$$

2.3 What does the set D mean?

A point $\mathbf{x} = (x_1, \ldots, x_n)$ belongs to D when its coordinates have product 1:

$$x_1 x_2 \cdots x_n = 1.$$

Although the definition says $x_k \geq 0$, the product condition automatically forces every coordinate to be strictly positive. Indeed, if one coordinate were 0, then the whole product would be 0, not 1. Thus every $\mathbf{x} \in D$ satisfies

$$x_k > 0 \qquad (k = 1, 2, \ldots, n).$$

For $n = 3$, some examples of points in D are

$$(1, 1, 1), \qquad \left(2, \frac{1}{2}, 1\right), \qquad \left(2, 2, \frac{1}{4}\right).$$

The coordinates may change, but their product must stay equal to 1.

It is useful to think of the x_k as *adjusting factors*. Making one x_k larger forces one or more of the others to become smaller so that the total product remains 1.

2.4 What does a minimum proof require?

Suppose we want to prove that the minimum of some expression is g. There are two separate jobs.

1. **Lower-bound job:** Show that every allowed choice gives a value at least g.
2. **Attainment job:** Find one allowed choice that gives exactly g.

The first job proves that the answer cannot be below g. The second proves that the answer is not above g. Together they prove that the minimum is exactly g.

3 The Main Tool: the AM–GM Inequality

For positive numbers $y_1, y_2, \ldots, y_n$, the arithmetic mean–geometric mean inequality says

$$\boxed{\frac{y_1 + y_2 + \cdots + y_n}{n} \geq (y_1 y_2 \cdots y_n)^{1/n}.} \tag{AM–GM}$$

Equality holds exactly when

$$y_1 = y_2 = \cdots = y_n.$$

For example,

$$\frac{2+8}{2} = 5 \geq \sqrt{2 \cdot 8} = 4.$$

The arithmetic mean is larger because the numbers 2 and 8 are unequal. If we use two equal numbers, such as 4 and 4, then

$$\frac{4+4}{2} = 4 = \sqrt{4 \cdot 4}.$$

A very important interpretation of AM–GM is this:

Among positive numbers with a fixed product, the sum is smallest when the numbers are all equal.

That sentence is exactly the idea behind Exercise 8.5.

4 Warm-Up: the Two-Number Case

When $n = 2$, the condition $x_1 x_2 = 1$ lets us write

$$x_1 = t, \qquad x_2 = \frac{1}{t}, \qquad t > 0.$$

The claimed formula becomes

$$\sqrt{a_1 a_2} = \min_{t>0} \frac{1}{2}\left(a_1 t + \frac{a_2}{t}\right).$$

Apply AM–GM to the two positive numbers

$$a_1 t \qquad \text{and} \qquad \frac{a_2}{t}.$$

We get

$$\begin{aligned} \frac{1}{2}\left(a_1 t + \frac{a_2}{t}\right) &\geq \sqrt{a_1 t \cdot \frac{a_2}{t}} \\ &= \sqrt{a_1 a_2}. \end{aligned}$$

The variable t disappears from the product because $t(1/t) = 1$.

Equality in AM–GM occurs when the two adjusted terms are equal:

$$a_1 t = \frac{a_2}{t}.$$

Therefore

$$a_1 t^2 = a_2, \qquad t = \sqrt{\frac{a_2}{a_1}}.$$

At this choice,

$$x_1 = \sqrt{\frac{a_2}{a_1}}, \qquad x_2 = \sqrt{\frac{a_1}{a_2}}.$$

Their product is 1, so this point is allowed.

4.1 A numerical example

Let

$$a_1 = 4, \qquad a_2 = 9.$$

The geometric mean is

$$\sqrt{4 \cdot 9} = 6.$$

The minimizing point is

$$x_1 = \sqrt{\frac{9}{4}} = \frac{3}{2}, \qquad x_2 = \sqrt{\frac{4}{9}} = \frac{2}{3}.$$

The two adjusted terms are equal:

$$a_1 x_1 = 4 \cdot \frac{3}{2} = 6, \qquad a_2 x_2 = 9 \cdot \frac{2}{3} = 6.$$

Hence

$$\frac{a_1x_1 + a_2x_2}{2} = \frac{6+6}{2} = 6.$$

The next figure gives a geometric picture. The curved line is the constraint $x_1x_2 = 1$. A straight line of the form

$$4x_1 + 9x_2 = c$$

contains points where the weighted sum has the same value. Moving such a line toward the origin makes c smaller. The smallest line that still touches the constraint curve is tangent at

$$\left(\frac{3}{2}, \frac{2}{3}\right).$$

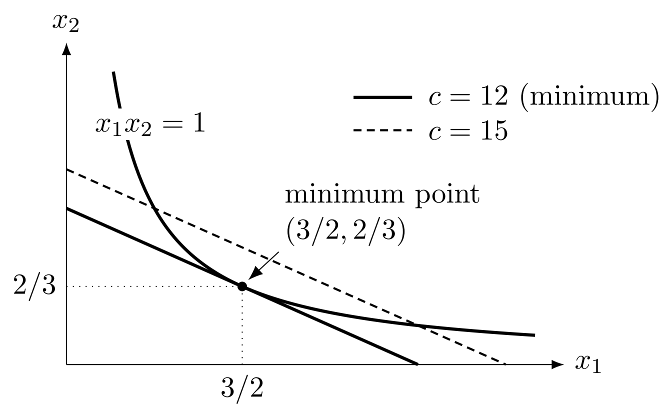

Figure 1: For $a_1 = 4$ and $a_2 = 9$, the smallest level line that meets $x_1x_2 = 1$ is $4x_1 + 9x_2 = 12$. Dividing by 2 gives the minimum value 6.

5 Complete Proof of the Minimum Representation

Let

$$g = \mathrm{GM}(\mathbf{a}) = \left(\prod_{k=1}^{n} a_k\right)^{1/n}.$$

We will carry out the two jobs described earlier.

5.1 Step 1: every allowed value is at least the geometric mean

Choose any point

$$\mathbf{x} = (x_1, x_2, \ldots, x_n) \in D.$$

Because $a_k > 0$ and $x_k > 0$, the numbers

$$a_1x_1, a_2x_2, \ldots, a_nx_n$$

are all positive. Apply AM–GM to them:

$$\begin{aligned}
\frac{1}{n}\sum_{k=1}^{n} a_k x_k &\geq \left(\prod_{k=1}^{n} a_k x_k\right)^{1/n} \\
&= \left(\left(\prod_{k=1}^{n} a_k\right)\left(\prod_{k=1}^{n} x_k\right)\right)^{1/n} \\
&= \left(\left(\prod_{k=1}^{n} a_k\right)\cdot 1\right)^{1/n} \\
&= g.
\end{aligned}$$

The third line used the defining property of D:

$$\prod_{k=1}^{n} x_k = 1.$$

This inequality is true for *every* point $\mathbf{x} \in D$. Therefore the smallest possible value cannot be less than g:

$$\min_{\mathbf{x}\in D} \frac{1}{n}\sum_{k=1}^{n} a_k x_k \geq g. \tag{1}$$

5.2 Step 2: construct a point that gives exactly the geometric mean

AM–GM becomes an equality when all its input numbers are equal. We therefore want

$$a_1 x_1 = a_2 x_2 = \cdots = a_n x_n = g.$$

This suggests choosing

$$\boxed{x_k^* = \frac{g}{a_k} \qquad (k = 1, 2, \ldots, n).} \tag{2}$$

Each x_k^* is positive. We must check that the product is 1:

$$\begin{aligned}
\prod_{k=1}^{n} x_k^* &= \prod_{k=1}^{n} \frac{g}{a_k} \\
&= \frac{g^n}{\prod_{k=1}^{n} a_k} \\
&= \frac{\prod_{k=1}^{n} a_k}{\prod_{k=1}^{n} a_k} \\
&= 1.
\end{aligned}$$

Here we used

$$g^n = \prod_{k=1}^{n} a_k.$$

Thus $\mathbf{x}^* \in D$.

Now evaluate the objective at this point:

$$\begin{aligned}\frac{1}{n}\sum_{k=1}^{n} a_k x_k^* &= \frac{1}{n}\sum_{k=1}^{n} a_k \frac{g}{a_k} \\ &= \frac{1}{n}\sum_{k=1}^{n} g \\ &= \frac{1}{n}(ng) \\ &= g.\end{aligned}$$

So an allowed point actually gives the value g. Therefore

$$\min_{\mathbf{x}\in D} \frac{1}{n}\sum_{k=1}^{n} a_k x_k \le g. \tag{3}$$

Combining (1) and (3), we obtain

$$\boxed{\left(\prod_{k=1}^{n} a_k\right)^{1/n} = \min_{\mathbf{x}\in D} \frac{1}{n}\sum_{k=1}^{n} a_k x_k.}$$

This proves formula (8.33).

5.3 What is the minimizing point?

The proof found the minimizer explicitly:

$$\boxed{\mathbf{x}^* = \left(\frac{g}{a_1}, \frac{g}{a_2}, \ldots, \frac{g}{a_n}\right).}$$

At this point,

$$a_1 x_1^* = a_2 x_2^* = \cdots = a_n x_n^* = g.$$

Thus the minimizing factors balance all the adjusted numbers perfectly.

Because equality in AM–GM requires all adjusted terms to be equal, this is the only minimizing point when all a_k are positive.

5.4 A three-number example

Take

$$\mathbf{a} = (1, 4, 16).$$

Its geometric mean is

$$g = (1 \cdot 4 \cdot 16)^{1/3} = 64^{1/3} = 4.$$

The minimizing factors are

$$\mathbf{x}^* = \left(\frac{4}{1}, \frac{4}{4}, \frac{4}{16}\right) = \left(4, 1, \frac{1}{4}\right).$$

Their product is

$$4 \cdot 1 \cdot \frac{1}{4} = 1,$$

and the adjusted numbers are

$$1 \cdot 4 = 4, \qquad 4 \cdot 1 = 4, \qquad 16 \cdot \frac{1}{4} = 4.$$

Consequently,

$$\frac{1 \cdot 4 + 4 \cdot 1 + 16 \cdot (1/4)}{3} = \frac{4 + 4 + 4}{3} = 4.$$

6 Using the Formula to Prove Superadditivity

Let

$$\mathbf{a} = (a_1, \ldots, a_n), \qquad \mathbf{b} = (b_1, \ldots, b_n),$$

with all entries positive. We want to prove

$$\mathrm{GM}(\mathbf{a} + \mathbf{b}) \geq \mathrm{GM}(\mathbf{a}) + \mathrm{GM}(\mathbf{b}).$$

6.1 A general fact about minima

Suppose F and H are two functions defined on the same set D. For every $x \in D$,

$$F(x) \geq \min_{u \in D} F(u) \qquad \text{and} \qquad H(x) \geq \min_{u \in D} H(u).$$

Adding gives

$$F(x) + H(x) \geq \min_{u \in D} F(u) + \min_{u \in D} H(u).$$

Because this is true for every $x \in D$, it remains true after minimizing the left-hand side:

$$\boxed{\min_{x \in D}\big(F(x) + H(x)\big) \geq \min_{x \in D} F(x) + \min_{x \in D} H(x).} \tag{4}$$

The direction $\geq$ is important. The function F may prefer one minimizing point, while H may prefer a different one. On the left, both functions must use the same point x.

6.2 Apply the general fact

For $\mathbf{x} \in D$, define

$$F_{\mathbf{a}}(\mathbf{x}) = \frac{1}{n} \sum_{k=1}^{n} a_k x_k, \qquad F_{\mathbf{b}}(\mathbf{x}) = \frac{1}{n} \sum_{k=1}^{n} b_k x_k.$$

Then

$$\begin{aligned} F_{\mathbf{a}+\mathbf{b}}(\mathbf{x}) &= \frac{1}{n} \sum_{k=1}^{n} (a_k + b_k) x_k \\ &= \frac{1}{n} \sum_{k=1}^{n} a_k x_k + \frac{1}{n} \sum_{k=1}^{n} b_k x_k \\ &= F_{\mathbf{a}}(\mathbf{x}) + F_{\mathbf{b}}(\mathbf{x}). \end{aligned}$$

Using the minimum representation (8.33) and then inequality (4),

$$\begin{aligned}
\mathrm{GM}(\mathbf{a}+\mathbf{b}) &= \min_{\mathbf{x}\in D} F_{\mathbf{a}+\mathbf{b}}(\mathbf{x}) \\
&= \min_{\mathbf{x}\in D}\big(F_{\mathbf{a}}(\mathbf{x}) + F_{\mathbf{b}}(\mathbf{x})\big) \\
&\geq \min_{\mathbf{x}\in D} F_{\mathbf{a}}(\mathbf{x}) + \min_{\mathbf{x}\in D} F_{\mathbf{b}}(\mathbf{x}) \\
&= \mathrm{GM}(\mathbf{a}) + \mathrm{GM}(\mathbf{b}).
\end{aligned}$$

Writing this without vector notation gives

$$\boxed{\left(\prod_{k=1}^{n}(a_k+b_k)\right)^{1/n} \geq \left(\prod_{k=1}^{n} a_k\right)^{1/n} + \left(\prod_{k=1}^{n} b_k\right)^{1/n}.}$$

This is the superadditivity of the geometric mean.

6.3 The same proof, written point by point

Some readers find the following version even clearer. Fix any $\mathbf{x} \in D$. Formula (8.33) says

$$\frac{1}{n}\sum_{k=1}^{n} a_k x_k \geq \mathrm{GM}(\mathbf{a})$$

and

$$\frac{1}{n}\sum_{k=1}^{n} b_k x_k \geq \mathrm{GM}(\mathbf{b}).$$

Adding,

$$\frac{1}{n}\sum_{k=1}^{n} (a_k+b_k) x_k \geq \mathrm{GM}(\mathbf{a}) + \mathrm{GM}(\mathbf{b}).$$

This is true for every $\mathbf{x} \in D$. Therefore it is true for the smallest possible left-hand side. But that smallest possible left-hand side is $\mathrm{GM}(\mathbf{a}+\mathbf{b})$ by (8.33). Hence

$$\mathrm{GM}(\mathbf{a}+\mathbf{b}) \geq \mathrm{GM}(\mathbf{a}) + \mathrm{GM}(\mathbf{b}).$$

6.4 A numerical check

Let

$$\mathbf{a} = (1, 4), \qquad \mathbf{b} = (9, 1).$$

Then

$$\mathrm{GM}(\mathbf{a}) = \sqrt{1\cdot 4} = 2, \qquad \mathrm{GM}(\mathbf{b}) = \sqrt{9\cdot 1} = 3.$$

Also,

$$\mathbf{a}+\mathbf{b} = (10, 5),$$

so

$$\mathrm{GM}(\mathbf{a}+\mathbf{b}) = \sqrt{10\cdot 5} = \sqrt{50} = 5\sqrt{2} \approx 7.07.$$

Thus

$$7.07 \geq 2 + 3 = 5.$$

The inequality is strict in this example.

6.5 When does equality occur?

For positive entries, equality occurs exactly when the two vectors are proportional. This means that there is a positive constant c such that

$$\mathbf{a} = c\mathbf{b},$$

or equivalently,

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \cdots = \frac{a_n}{b_n}.$$

Here is the reason. The minimizing point for $\mathbf{a}$ is

$$x_k^{(a)} = \frac{\mathrm{GM}(\mathbf{a})}{a_k},$$

while the minimizing point for $\mathbf{b}$ is

$$x_k^{(b)} = \frac{\mathrm{GM}(\mathbf{b})}{b_k}.$$

For equality in the superadditivity proof, the same point must minimize both expressions. Therefore

$$\frac{\mathrm{GM}(\mathbf{a})}{a_k} = \frac{\mathrm{GM}(\mathbf{b})}{b_k} \qquad \text{for every } k,$$

which says that a_k/b_k is independent of k.

For example, take

$$\mathbf{a} = (1, 4), \qquad \mathbf{b} = (2, 8) = 2\mathbf{a}.$$

Then

$$\mathrm{GM}(\mathbf{a}) = 2, \qquad \mathrm{GM}(\mathbf{b}) = 4,$$

and

$$\mathrm{GM}(\mathbf{a} + \mathbf{b}) = \mathrm{GM}(3, 12) = \sqrt{36} = 6 = 2 + 4.$$

7 Why the Minimum Representation Is So Useful

The original geometric mean

$$\left(\prod_{k=1}^{n} a_k\right)^{1/n}$$

contains a product and an nth root. Those operations can be difficult to handle when we add two vectors.

Formula (8.33) rewrites the same quantity as a minimum of expressions that are linear in the entries a_k:

$$\frac{1}{n}\sum_{k=1}^{n} a_k x_k.$$

Linear expressions split perfectly over addition:

$$\sum_{k=1}^{n} (a_k + b_k)x_k = \sum_{k=1}^{n} a_k x_k + \sum_{k=1}^{n} b_k x_k.$$

That simple splitting is the entire reason the superadditivity proof becomes short.

A useful summary is

$$\boxed{\text{fixed product constraint } + \text{ AM–GM} \implies \text{minimum formula} \implies \text{superadditivity.}}$$

8 Technical Note About Zero Entries

The exercise writes "minimum." This is correct when all $a_k > 0$. If some, but not all, of the a_k are zero, then the geometric mean is 0, but the minimum on the right usually is not attained.

For the simplest example, take $n = 2$ and

$$(a_1, a_2) = (0, 1).$$

Every point in D has the form $(t, 1/t)$ with $t > 0$, and

$$\frac{1}{2}\left(a_1 t + a_2 \frac{1}{t}\right) = \frac{1}{2t}.$$

As t becomes larger and larger,

$$\frac{1}{2t} \longrightarrow 0.$$

The values can get arbitrarily close to 0, but they never equal 0 for a finite positive t. Therefore there is no minimum; there is only an *infimum*, meaning the greatest lower bound.

For nonnegative inputs, the completely general formula is

$$\boxed{\left(\prod_{k=1}^{n} a_k\right)^{1/n} = \inf_{\mathbf{x} \in D} \frac{1}{n} \sum_{k=1}^{n} a_k x_k.}$$

When all $a_k > 0$, the infimum is attained at $x_k = g/a_k$, so it is an actual minimum. The superadditivity proof works with inf in exactly the same way. One may also prove the positive case first and then approach zero entries by a limit.

9 Common Mistakes to Avoid

- **Forgetting that the product constraint matters.** The cancellation

$$\prod_{k=1}^{n} a_k x_k = \left(\prod_{k=1}^{n} a_k\right)\left(\prod_{k=1}^{n} x_k\right) = \prod_{k=1}^{n} a_k$$

  works only because $\prod x_k = 1$.

- **Proving only a lower bound.** Showing that every value is at least g does not by itself prove that the minimum equals g. One must also exhibit the point $x_k = g/a_k$.

- **Choosing the factors in the wrong direction.** Large a_k need small x_k, and small a_k need large x_k. The correct choice is $x_k = g/a_k$, not $x_k = a_k/g$.

- **Reversing the minimum inequality.** In general,

$$\min(F+H) \geq \min F + \min H,$$

not the other way around.

- **Thinking that $x_k = 0$ is allowed.** The notation says $x_k \geq 0$, but the condition $\prod x_k = 1$ rules out zero.

- **Ignoring the issue of zero coefficients.** If some $a_k = 0$, the general statement uses an infimum unless the surrounding text has already assumed all $a_k > 0$.

10 The Two Proofs in Compact Form

Minimum representation. For every $\mathbf{x} \in D$, AM–GM gives

$$\frac{1}{n}\sum_{k=1}^{n} a_k x_k \geq \left(\prod_{k=1}^{n} a_k x_k\right)^{1/n} = \left(\prod_{k=1}^{n} a_k\right)^{1/n}.$$

Equality is obtained by choosing

$$x_k = \frac{(\prod_{j=1}^{n} a_j)^{1/n}}{a_k}.$$

Therefore (8.33) holds.

Superadditivity. For every $\mathbf{x} \in D$,

$$\begin{aligned}\frac{1}{n}\sum_{k=1}^{n}(a_k+b_k)x_k &= \frac{1}{n}\sum_{k=1}^{n} a_k x_k + \frac{1}{n}\sum_{k=1}^{n} b_k x_k \\ &\geq \mathrm{GM}(\mathbf{a}) + \mathrm{GM}(\mathbf{b}).\end{aligned}$$

Taking the minimum over D on the left gives

$$\mathrm{GM}(\mathbf{a}+\mathbf{b}) \geq \mathrm{GM}(\mathbf{a}) + \mathrm{GM}(\mathbf{b}).$$

Quick Self-Check

1. Why is every coordinate of a point in D actually positive?
   Answer: If any coordinate were zero, the product of all coordinates would be zero instead of one.

2. Which n positive numbers receive AM–GM in the main proof?
   Answer: The adjusted numbers $a_1x_1, a_2x_2, \ldots, a_nx_n$.

3. Why does their product not depend on the chosen point $\mathbf{x} \in D$?
   Answer:

   $$\prod(a_k x_k) = (\prod a_k)(\prod x_k) = \prod a_k$$

   because $\prod x_k = 1$.

4. What choice makes equality hold in AM–GM?
   Answer:
$$x_k = \frac{g}{a_k}, \qquad g = (\prod a_j)^{1/n},$$
   so that every adjusted term $a_k x_k$ equals g.
5. Why is $\min(F+H)$ usually not equal to $\min F + \min H$?
   Answer: The two functions may achieve their separate minima at different points, while $F+H$ must use one common point.
6. What does superadditivity say in words?
   Answer: The geometric mean of the coordinatewise sum is at least the sum of the two separate geometric means.